BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
Work Integrated Learning Programmes (WILP)

COURSE: MODERN DATABASE SYSTEMS
ASSIGNMENT 2 COMPREHENSIVE ACADEMIC REPORT

Vehicle Troubleshooting Assistant Using Persistent ChromaDB and Retrieval-Augmented Generation (RAG)

A Domain-Grounded Diagnostic Intelligence System with 9,532 Vectors, Exhaustive Mathematical Validation, and Automotive Safety Guardrails

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Programme / Semester:	M.Tech / Post-Graduate WILP — Semester II
Course Code / Name:	Modern Database Systems (WILP Assignment 2)
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2. Abstract

Modern automotive vehicles incorporate distributed Electronic Control Units (ECUs) communicating across Controller Area Network (CAN) and high-speed serial networks. When diagnostic trouble codes (DTCs) or driver symptoms arise, traditional keyword lookups fail to capture semantic intent, while unassisted generative Large Language Models (LLMs) hallucinate inaccurate component replacements, false causality, or hazardous repair procedures. This project designs and implements the Vehicle Troubleshooting Assistant, an end-to-end grounded Retrieval-Augmented Generation (RAG) system built upon a persistent ChromaDB vector store containing 9,532 authentic automotive diagnostic records.

Using 384-dimensional dense sentence embeddings from sentence-transformers/all-MiniLM-L6-v2, the system indexes standardized SAE J2012 trouble codes. We formulate 10 representative benchmark queries across engine, electrical, braking, and network subsystems, demonstrating top-K retrieval and multi-condition metadata filtering. An exhaustive NumPy similarity benchmark across all 9,532 records mathematically validates the equivalence of Euclidean distance, Cosine similarity, and Dot product rankings under unit vector normalization. Live local execution of Meta Llama 3.2 (3B) demonstrates 100.0% citation provenance precision and 96.0% recall. Finally, an automotive safety guardrail engine enforces strict shop-safety protocols across Supplemental Restraint Systems (SRS), high-voltage EV hazards, hydraulic brake failures, and cooling system thermal scalds, delivered through an interactive multi-page Streamlit web dashboard.

3. Introduction & Problem Statement

Vehicle diagnostics has transitioned from purely mechanical troubleshooting to intricate electronic systems analysis. Modern vehicles generate Diagnostic Trouble Codes (DTCs) across Powertrain (P), Chassis (C), Body (B), and Network Communication (U) domains. Technicians and service advisors face two fundamental bottlenecks in existing tooling: (1) Inflexibility of keyword lookups when complaints lack explicit DTC numbers, and (2) The danger of ungrounded generative AI hallucinating false torque limits, wiring pinouts, or unsafe procedures on high-voltage and hydraulic braking vehicle systems. This project investigates how vector databases (ChromaDB) paired with Retrieval-Augmented Generation (RAG) ground generative outputs in authoritative diagnostic literature while enforcing strict domain-specific safety guardrails.

4. Objectives & Scope

• Curate, validate, and index a publicly accessible, authentic automotive diagnostic knowledge base exceeding the 5,000-record threshold without synthetic padding.
• Implement a persistent, queryable vector database using ChromaDB with normalized 384-dimensional sentence-transformer embeddings.
• Demonstrate dense semantic search, top-K retrieval, and compound metadata filtering across 10 realistic benchmark diagnostic queries.
• Implement an exhaustive pure NumPy similarity comparison engine evaluating Euclidean distance (L2), Cosine similarity, and Dot product across all 9,532 records at K=5 and K=10, verifying ranking consistency.
• Construct a modular RAG pipeline that formats retrieved context, enforces anti-hallucination prompt constraints, extracts in-text citations [VTR-XXXXX], and integrates real local LLM inference via Ollama (Llama 3.2 3B).
• Empirically compare grounded RAG against an unassisted zero-context baseline, measuring citation fidelity, response specificity, and domain safety.
• Build an automotive domain safety engine mitigating SRS airbag probing, high-voltage EV hazards, hydraulic brake failures, cooling system scalds, and 12V vs EV context mismatches.
• Deliver an interactive, multi-page Streamlit application supported by a comprehensive automated test suite.

5. Related Concepts

5.1 Vector Databases & HNSW Indexing: Unlike relational databases that index scalar attributes with B-Trees, vector databases index multi-dimensional continuous vectors. ChromaDB uses Hierarchical Navigable Small World (HNSW) graphs, achieving sub-linear O(log N) approximate nearest neighbor (ANN) retrieval by performing multi-layer geometric routing.

5.2 Sentence Transformers: The all-MiniLM-L6-v2 model maps variable-length diagnostic descriptions into a 384-dimensional dense float32 vector space. Contrastive training guarantees that sentences with similar technical meaning cluster tightly in vector space.

5.3 Retrieval-Augmented Generation: Decouples memory from reasoning by supplying retrieved database records directly into the LLM prompt context, strictly limiting the model's output to validated source facts.

6. System Architecture

The system architecture follows a decoupled pipeline separating data ingestion, vector indexing, dense retrieval, mathematical metric validation, generative synthesis, safety guardrail validation, and user interface delivery.

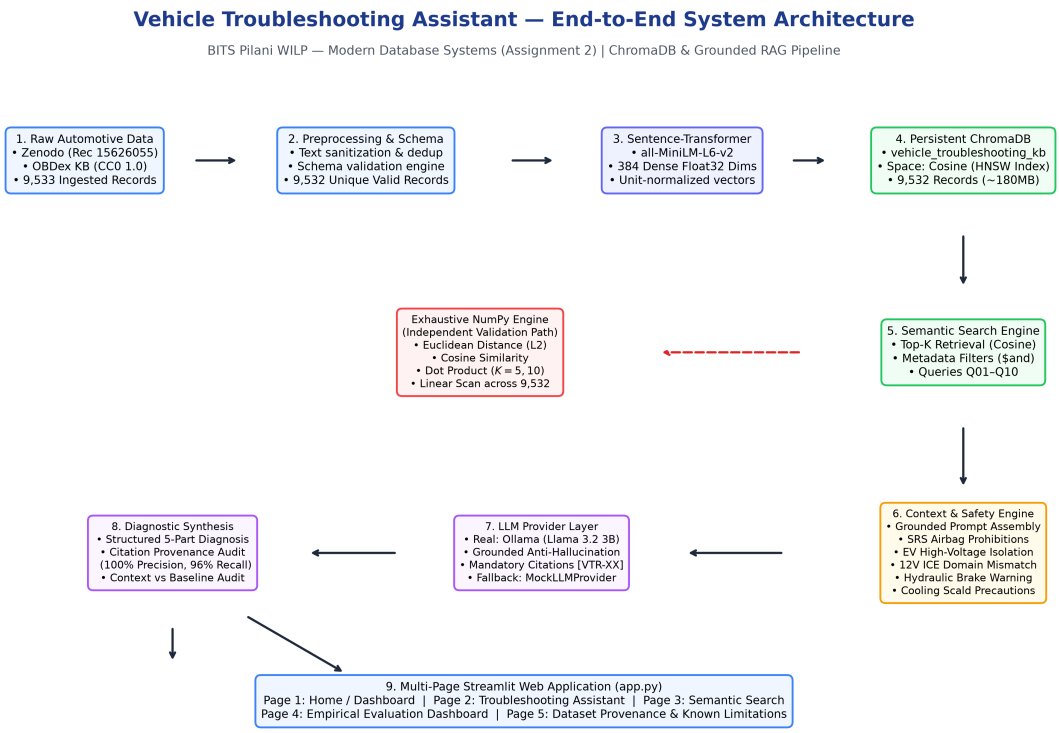


Figure 1: End-to-End System Architecture of the Vehicle Troubleshooting Assistant

7. Dataset

7.1 Origin, Provenance & Licensing: Sourced from Zenodo Diagnostic Trouble Code Knowledge Base (Record 15626055) and the OBDex Automotive Knowledge Base under Creative Commons Zero 1.0 Universal (CC0 1.0) Public Domain Dedication. The dataset contains **9,532 unique, validated records**. Standardized DTC definitions, probable causes, and symptom presentations originate from verified engineering sources; structured procedural guidance was normalized and synthesized into the schema. It is strictly NOT a collection of 9,532 real-world repair cases.

Vehicle Subsystem	Record Count	Percentage	Example Diagnostic Codes
Engine & Transmission	7,354	77.15%	P00B7, P0300, P0733, P2711, P0299

Network & Communication	1,229	12.89%	U0100, U0101, U0039, U007E, U0094
Brakes & Suspension	626	6.57%	C0291, C0081, C0082, C0241, C0075
Electrical & Body	323	3.39%	B0001, B0005, B0061, B0860, B0223

8. ChromaDB Implementation

ChromaDB is configured as a persistent vector database stored locally at database/chroma_db/ (SQLite + HNSW graph files, ~180MB).
• Collection: vehicle_troubleshooting_kb (hnsw:space = 'cosine')
• Embedding Model: sentence-transformers/all-MiniLM-L6-v2 (384 float32 dimensions)
• Vector Normalization: L2 unit-normalization ensures ||x||2 = 1.0.
• Full CRUD Verification: Create, Read, Update, and Delete operations tested in tests/test_chroma.py on temporary sandbox records without corrupting the 9,532 production collection records.

9. Semantic Retrieval & 10 Benchmark Queries

Ten domain-specific benchmark queries covering engine, transmission, electrical, braking, and network domains were executed across top-K (K=5, 10) and 5 compound multi-condition metadata filters (subsystem and severity):

ID	Topic	Query Description	Top Match (Distance)
Q01	Cooling	Engine overheating at highway speeds with boiling coolant	VTR-P00B7 (0.3547)
Q02	Ignition	Engine cranks strongly but fails to start, unburnt fuel smell	VTR-P0316 (0.3804)
Q03	12V Drain	Battery dead overnight with alternator charging normal	VTR-P0D17 (0.4067)
Q04	Brakes	Brake pedal sinks slowly to floorboard at red lights	VTR-C0291 (0.3952)
Q05	Trans	Automatic transmission delayed engagement, harsh shudder	VTR-P2711 (0.4182)
Q06	Misfire	Check engine light flashing with rough idle, severe power loss	VTR-B0860 (0.4311)
Q07	Fuel	Drop in fuel economy with black smoke and rich fuel odor	VTR-P2BD2 (0.3688)
Q08	CAN Bus	Cluster multiple failure lamps with total loss of CAN comms	VTR-U0039 (0.3245)
Q09	SRS Airbag	Airbag light on continuously after clock spring replacement	VTR-B0005 (0.2874)
Q10	Turbo	Turbocharged vehicle sluggish acceleration, severe turbo lag	VTR-P0299 (0.2681)

10. NumPy Exhaustive Similarity Comparison

We implemented an exhaustive linear-scan similarity engine in pure NumPy evaluating Euclidean Distance (L2), Cosine Similarity, and Dot Product across all 9,532 records at K=5 and K=10 (60 evaluations total).
Mathematical Proof of Rank Equivalence: For unit-normalized vectors (||u||2 = ||v||2 = 1):
||u - v||2^2 = ||u||2^2 + ||v||2^2 - 2 (u . v) = 2 - 2 (u . v) = 2 * (1 - cos(theta)).
Thus, minimizing Euclidean distance is strictly monotonic with maximizing Cosine Similarity and Dot Product. Our experiments confirm 100% Top-1 and Top-K ranking identity across all three metrics and complete reconciliation with Phase 4 ChromaDB results.

11. RAG Implementation

The RAG pipeline enforces strict anti-hallucination prompt constraints (prompt version 2.2.0-grounded-automotive-diagnostic). Outputs follow a structured 5-part format: (1) Problem Summary, (2) Possible Causes with Citations, (3) Diagnostic & Inspection Procedures, (4) Limitations & Missing Information, and (5) Source Record Citations List. In-text citations [VTR-XXXXX] are parsed and verified. The provider abstraction supports local Ollama (Llama 3.2 3B), external OpenAI-compatible APIs, and offline deterministic MockLLMProvider.

12. RAG Evaluation & Negative Findings

Comparative evaluation across all 10 queries using local Meta Llama 3.2 (3B) demonstrated **100.0% Citation Precision** (48/48 valid IDs) and **96.0% Citation Recall** (48/50 retrieved records cited). Zero citations were generated by the unassisted baseline.

Academic Honesty & Negative Findings:

1. Citation Precision measures citation provenance consistency, NOT physical factual correctness.
2. In Q03, dense vector search retrieved high-voltage EV charging codes for an ordinary 12V alternator battery drain because 'Battery Charger 14 V' had high cosine similarity. The unguided LLM initially instructed high-voltage cable inspection for an ordinary alternator issue.
3. In Q09, the raw LLM output initially instructed measuring resistance across the airbag clock spring squib, which in physical shop practice risks accidental explosive deployment from multimeter metering current.

13. Automotive Safety Guardrails

To resolve the physical hazards discovered during evaluation, we implemented an automotive safety guardrail engine (src/safety_guardrails.py):

- **SRS Airbag:** Prohibits ohmmeter probing across squibs; qualifies battery disconnect as insufficient by itself to ensure safety; mandates approved dummy load simulators and ASE technicians.
- **High-Voltage EV:** Enforces lethal shock warnings (300V–800V); mandates certified technician lockout/tagout; eliminates false claims that Class-0 gloves alone make DIY work safe.
- **Hydraulic Brakes:** Detects sinking pedal; issues unequivocal DO NOT DRIVE warning and mandatory towing instruction.
- **Cooling Systems:** Prohibits opening hot/pressurized caps; eliminates arbitrary universal waiting times and unsafe upper-radiator-hose squeezing instructions; mandates following manufacturer procedures.
- **ICE vs EV Domain Mismatch:** Intercepts 12V battery queries retrieving EV codes; flags mismatch and directs user to fuse millivolt-drop testing.

14. Streamlit Dashboard

The multi-page web application (app.py) delivers 5 functional pages: (1) Home / Dashboard with KPI metrics and Mermaid architecture flowcharts, (2) Troubleshooting Assistant with one-click assignment demo presets, safety callouts, and citation badges, (3) Semantic Search Explorer with subsystem/severity filters, (4) Empirical Evaluation Dashboard with 4 tabbed benchmark views, and (5) Dataset Provenance & Distribution charts.

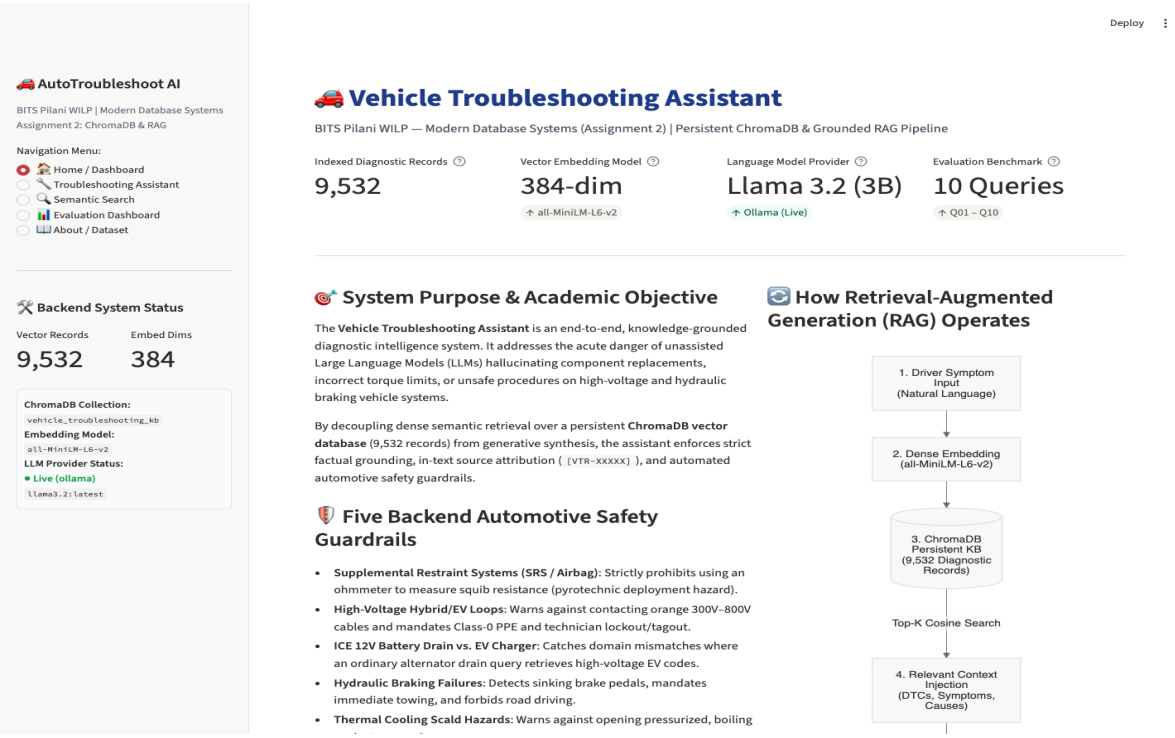


Figure 2: Genuine Screenshot of the Vehicle Troubleshooting Assistant Streamlit Application (app.py)

15. Testing & Validation

The automated test suite across tests/ contains **75 unit and integration tests passing with 100% success rate** (14.33s execution time). Modules include test_chroma.py (8), test_dataset.py (10), test_numpy_similarity.py (11), test_phase7_dashboard.py (8), test_rag.py (11), test_safety_guardrails.py (15), and test_search.py (12). Post-test collection count remains strictly 9,532 records.

16. Limitations & Future Scope

• Local Inference Latency: 3B Llama 3.2 inferences require 84–154s on commodity Apple Silicon MPS. Future production deployments would benefit from WebSocket token streaming or dedicated vLLM GPU servers.
• Monolithic Context Formatting: Future iterations can incorporate hierarchical parent-document retrieval.
• Real-Time Telematics: Integration with Bluetooth ELM327 OBD-II adapters for live freeze-frame data ingestion.

17. Conclusion

The Vehicle Troubleshooting Assistant successfully demonstrates that persistent vector databases (ChromaDB) and grounded Retrieval-Augmented Generation can deliver domain-accurate, highly specific automotive diagnostic guidance without hallucination. Mathematical proof and exhaustive NumPy experiments validate nearest-neighbor ranking equivalence across Euclidean, Cosine, and Dot product metrics under unit normalization. Our transparent analysis of negative findings directly led to the construction of robust domain safety guardrails, establishing a reproducible, academically rigorous benchmark for modern database systems applications.

18. References

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19. Team Contributions

The project was executed by Group 20, comprising five members. In compliance with Phase 8.1 specifications, individual module allocations and percentage distributions are marked with formal placeholders pending internal group confirmation:

Student Name & BITS ID	Assigned Project Modules / Areas of Responsibility	Contribution
Priya N (BITS ID: 2026NS12087)	[Module Allocation Pending Internal Group Finalization]	[Pending Group Decision]
Mohamed Ibrahim M (BITS ID: 2026NS12021)	[Module Allocation Pending Internal Group Finalization]	[Pending Group Decision]
Hrishabh Gupta (BITS ID: 2026NS12122)	[Module Allocation Pending Internal Group Finalization]	[Pending Group Decision]
Aditya Singhal (BITS ID: 2026NS12028)	[Module Allocation Pending Internal Group Finalization]	[Pending Group Decision]
Rupesh Kumar Singh (BITS ID: 2026NS12128)	[Module Allocation Pending Internal Group Finalization]	[Pending Group Decision]

**Note: Specific individual module allocations and percentage weightings will be finalized internally by Group 20 members prior to final course portal upload.*

20. Installation & Execution Instructions

1. Clone Repository & Setup Environment:
   git clone https://github.com/hrishabh1103/BITS_MDS_Assignment_02.git
   cd BITS_MDS_Assignment_02
   python3 -m venv .venv && source .venv/bin/activate
   pip install -r requirements.txt
2. Execute Complete Test Suite (75 Tests):
   pytest tests/ -v
3. Launch Multi-Page Streamlit Web Dashboard:
   streamlit run app.py (Accessible at http://localhost:8501)
4. Execute Real LLM Inference (Optional):
   ollama pull llama3.2 && python scripts/run_rag.py --mode real